

Indian agriculture needs to adapt better to climate change

The Indian farming **ecosystem** has seen a big change in recent years — the acceptance of climate change as an **imminent** concern, rather than an issue that needs to be tackled in the medium term. Policymakers, bankers, researchers, weather experts and **seed companies** are of the view that cropping techniques, patterns and technologies must adapt rapidly.

At the recently **concluded** businessline Agri and Commodity **summit**, it was agreed that 'natural farming' can **insulate** India's crops against increasingly frequent extreme weather events. If the Indian Council for Agricultural Research is working on **drought** and flood **resistant** varieties, Nabard is promoting new financial **mechanisms** such as the Agri Fund and the Carbon Fund, working **in tandem with** States and **multilateral** agencies. **In terms of** the delivery of climate-proofing solutions, primary agricultural credit societies have been identified as important **nodal** institutions. The Centre plans to **expand** their role beyond giving credit. However, the **drive** to **set up** two lakh multi-purpose cooperatives **would** require the cooperation of the States. There is growing **recognition** of the **potential** of earning **carbon credits** through climate-smart practices, although the processes need to be **firmed up**, so that genuine farmers rather than **dubious** 'consultants' and 'experts' with global links benefit. **Funding** for such practices **must** be **accompanied** by clearcut conditions, so that back-end inputs too are 'green'.

However, there are gaps in the current **thrust**. India's farm research systems are primarily **geared towards** increasing **yields** through resource-intensive hybrid varieties. It is not clear whether varieties that combine stress **resistance** and yield improvement are **in the pipeline**. More efforts should be made to select natural varieties for their **resilient traits**. A research **focus** on yields alone **may** prove **limiting in the long run** as the **prospect** of crop damage **assumes** more serious proportions. The Prime Minister's Office has **called for** reforms in the ICAR system. While foodgrain output has increased despite **vagaries** of the weather, it is better to be prepared for **eventualities**.

It is also important to realise that the **impact** of harsh weather on crops **is pronounced** in the **rainfed heartland**, which **constitutes** 50 per cent of the cultivated area, as against irrigated regions. Climate proofing policies should focus on the unirrigated regions **in particular**. **Multi-cropping** as well as promotion of millets on typically small **patches** of land **can** protect output and incomes. Research has shown that monocropping in rainfed regions has reduced returns to farmers, while **exposing** them to uncertainty. There needs to be more conceptual clarity on the entire **gamut** of processes that **encompass sustainable** farming. The **allocation** towards the PM Paramparagat Krishi Vikas Yojana, which has seen just over ₹2,000 crore being spent over about four years, **can** be increased alongside reforms in research and **market linkages**.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

1. **Adapt** (verb) – Adjust, modify, alter, change अनुकूलित करना
2. **Ecosystem** (noun) – System, environment, community, framework पारिस्थितिकी तंत्र
3. **Imminent** (adjective) – About to happen, forthcoming, impending, close at hand आसन्न
4. **Seed company** (noun) – Enterprise dealing in seeds, seed-seller, seed producer बीज कंपनी
5. **Concluded** (adjective) – Finished, ended, final, completed समाप्त
6. **Summit** (noun) – Conference, meeting, gathering, conclave शिखर सम्मेलन
7. **Insulate** (verb) – Protect, shield, safeguard, cushion बचाव करना
8. **Drought** (noun) – Dry spell, water shortage, aridity, prolonged lack of rain सूखा
9. **Resistant** (adjective) – Able to withstand, impervious, resilient, proof against प्रतिरोधी
10. **Mechanism** (noun) – Method, process, system, apparatus तंत्र
11. **In tandem with** (phrase) – Together with, in conjunction with, alongside के साथ मिलकर
12. **Multilateral** (adjective) – Involving multiple parties, many-sided, cooperative बहुपक्षीय
13. **In terms of** (phrase) – Regarding, considering, from the perspective of के संदर्भ में
14. **Nodal** (adjective) – Central, pivotal, focal, main प्रमुख
15. **Expand** (verb) – Enlarge, extend, broaden, widen विस्तार करना
16. **Drive** (noun) – Campaign, push, initiative, effort अभियान
17. **Set up** (phrasal verb) – Establish, start, found, form स्थापित करना
18. **Recognition** (noun) – Acknowledgement, acceptance, appreciation, awareness मान्यता
19. **Potential** (noun) – Possibility, capability, capacity, promise संभावना
20. **Carbon credit** (noun) – Certificate offsetting emissions, tradeable permit for CO₂ reduction कार्बन क्रेडिट
21. **Firm up** (phrasal verb) – Make more solid, finalize, strengthen, confirm सुदृढ़ करना
22. **Dubious** (adjective) – Suspicious, questionable, unsure, doubtful संदिग्ध

23. **Accompany** (verb) – Go along with, occur with, come with, attend साथ होना
24. **Thrust** (noun) – Main point, focus, emphasis, drive मुख्य बल/उद्देश्य
25. **Gear towards** (phrasal verb) – Tailor to, direct toward, orient towards, focus on केंद्रित होना
26. **Yield** (noun) – Produce, output, harvest, amount उपज
27. **Resistance** (noun) – Opposition, resilience, immunity, durability प्रतिरोध
28. **In the pipeline** (phrase) – In progress, on the way, under development विकासाधीन
29. **Resilient** (adjective) – Tough, flexible, enduring, adaptable मजबूत
30. **Trait** (noun) – Characteristic, quality, feature, attribute गुण
31. **Limit** (verb) – Restrict, confine, curb, constrain सीमित करना
32. **In the long run** (phrase) – Over time, eventually, ultimately, in the end लंबे समय में
33. **Prospect** (noun) – Possibility, likelihood, potential outcome, future scenario संभावना
34. **Assume** (verb) – Take on, presume, accept, adopt मान लेना/ग्रहण करना
35. **Call for** (phrasal verb) – Require, demand, need, necessitate आवश्यकता होना
36. **Vagaries** (noun) – Uncertainties, changes, fluctuations, peculiarities अनिश्चितताएँ
37. **Eventuality** (noun) – Possible outcome, contingency, circumstance, future possibility संभावित स्थिति
38. **Pronounced** (adjective) – Noticeable, marked, evident, strong स्पष्ट/उच्च स्तर का
39. **Rainfed** (adjective) – Dependent on rainfall, relying on rain पानी पर निर्भर
40. **Heartland** (noun) – Core area, central region, main region केंद्र क्षेत्र
41. **Constitute** (verb) – Make up, form, comprise, represent होना/निर्मित करना
42. **In particular** (phrase) – Specifically, especially, notably विशेष रूप से
43. **Patch** (noun) – Small piece of land, plot, area, segment छोटा क्षेत्र
44. **Expose** (verb) – Make vulnerable, reveal, subject to, uncover उजागर करना
45. **Gamut** (noun) – Range, scope, extent, span व्यापक स्पेक्ट्रम
46. **Encompass** (verb) – Include, cover, embrace, span सम्मिलित करना

47. **Sustainable** (adjective) – Environmentally friendly, long-term, viable, renewable
टिकाऊ

48. **Allocation** (noun) – Distribution, apportionment, allotment, assignment
आवंटन

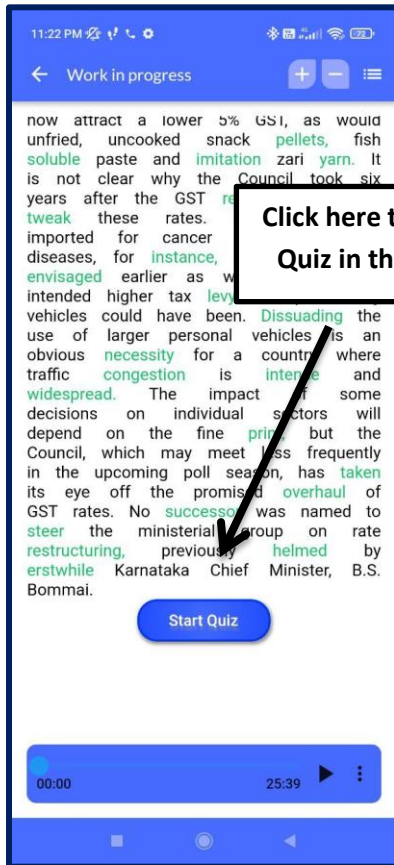
49. **Market linkage** (noun) – Connection between producers and markets, channel for sales बाज़ार से जुड़ाव

@mocks_wallah

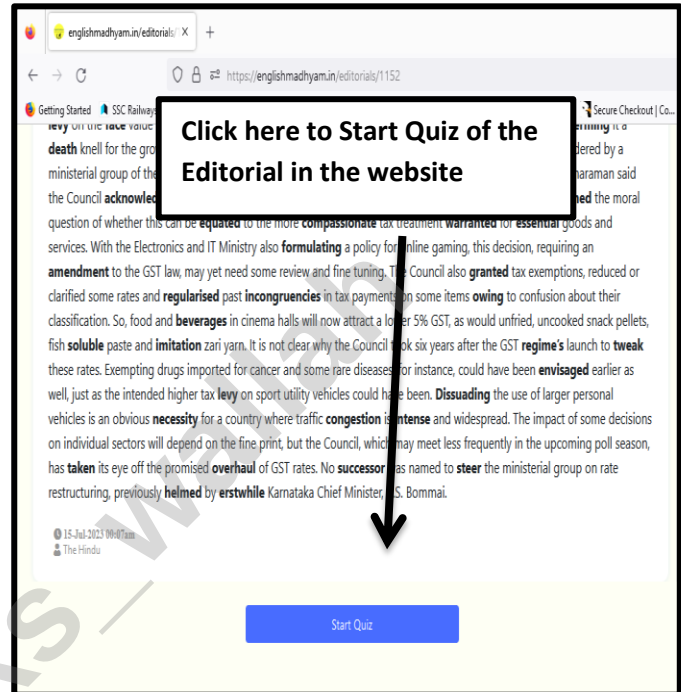
Summary of the Editorial

1. **Recognition of Climate Change:** Indian agriculture stakeholders now recognize climate change as an imminent concern, requiring urgent action.
2. **Adaptation Focus:** Cropping techniques, patterns, and technologies must adapt rapidly to changing climatic conditions.
3. **Natural Farming's Role:** Natural farming is viewed as a key method to protect crops against extreme weather events.
4. **Research on Resistant Varieties:** The Indian Council for Agricultural Research (ICAR) is developing drought and flood-resistant crop varieties.
5. **Innovative Financial Mechanisms:** Nabard is promoting new funding mechanisms like the Agri Fund and Carbon Fund in partnership with states and multilateral agencies.
6. **Primary Agricultural Credit Societies:** These societies are pivotal in delivering climate-proofing solutions, with plans to expand their roles beyond providing credit.
7. **Cooperative Expansion:** Setting up two lakh multi-purpose cooperatives requires close coordination with state governments.
8. **Carbon Credits Potential:** Climate-smart practices could help farmers earn carbon credits, but processes must ensure that genuine farmers benefit.
9. **Resource Efficiency:** Funding for green farming must enforce conditions ensuring sustainable back-end inputs.
10. **Research Gaps:** Current farm research prioritizes yield increases but lacks a focus on combining stress resistance and yield improvement.
11. **Resilient Varieties:** Emphasis should shift to selecting natural varieties for their resilience against climate stress.
12. **Rainfed Regions at Risk:** Unirrigated regions, which form 50% of the cultivated area, are most vulnerable to harsh weather impacts.
13. **Multi-Cropping and Millets:** Promoting multi-cropping and millet cultivation can enhance resilience and protect farmer incomes in rainfed areas.
14. **Monocropping Risks:** Monocropping in rainfed areas reduces farmer returns and increases vulnerability to uncertainty.
15. **Policy and Resource Allocation:** Increased funding for sustainable farming programs like PM Paramparagat Krishi Vikas Yojana, coupled with research and market reforms, is essential for long-term climate adaptation.

Practice Exercise: Banking Pattern Based



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